Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel International Advanced Level

Centre Number | Candidate Number

Wednesday 30 October 2019

Afternoon (Time: 1 hour 30 minutes) | Paper Reference **WST02/01**

Mathematics

International Advanced Subsidiary/Advanced Level
Statistics S2

You must have:
Mathematical Formulae and Statistical Tables (Blue), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions
- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of the tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information
- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice
- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ▶

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1. The proportion of people in a large population who wear glasses is p.
A random sample of 4 people is taken from the population.
The random variable X represents the number of people in the sample who wear glasses.

(a) Show that $P(X > 2) = p^3(4 - 3p)$

(3)

Given that more than half of the population wear glasses and that the standard deviation of X is 0.96

(b) find $P(X > 2)$

(4)

Given also that more than 2 people in the sample wear glasses,

(c) find the probability that exactly 3 people in the sample wear glasses.

(2)

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Question 1 continued

Q1

(Total 9 marks)

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2. The continuous random variable X is uniformly distributed over the interval $[-3, 12]$

(a) Find the upper quartile of X.

(2)

(b) Find $P(5 \leqslant X < 14)$

(2)

The continuous random variable Y is uniformly distributed over the interval $[a, b]$ where $0 < a < b$

Given that $E(X) = E(Y - 3)$ and $Var(X) = 4Var(Y)$

(c) find the value of a and the value of b.

(5)

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Question 2 continued

Q2

(Total 9 marks)

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3. For a particular website, hacking attempts occur randomly at a rate of 0.3 per hour.

(a) Show that the probability of at least one hacking attempt taking place in a randomly selected one-hour period is 0.259 to 3 significant figures.

(2)

(b) Find the probability that exactly 6 hacking attempts take place in a randomly selected 24-hour period.

(3)

Following a security update, Saira believes that the rate of hacking attempts on the website has decreased. She records 38 hacking attempts in a 7-day period following the update.

(c) Comment on Saira's belief by using an appropriate approximation and a suitable test at the 5% level of significance. State your hypotheses clearly.

(7)

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Question 3 continued

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Question 3 continued

(Total 12 marks)

Q3

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4. The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{15}x^2(3-x) & 1 \leqslant x < 3 \\ \frac{3}{10}(x-3) & 3 \leqslant x \leqslant 5 \\ 0 & \text{otherwise} \end{cases}$$

(a) Use algebraic integration to find $E(X^2)$

(4)

(b) Define fully the cumulative distribution function $F(x)$

(5)

(c) Find $P(2 < X < 4)$

(2)

(d) Find, to 3 significant figures, the value of k such that $P(X > k) = 0.2$

(3)

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Question 4 continued

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Question 4 continued

(Total 14 marks)

Q4

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5. A bag contains a large number of 1p coins and 2p coins in the ratio 2 : 3 respectively.

A jar contains a large number of 2p coins and 5p coins in the ratio 1 : 4 respectively.

Two coins are selected at random from the bag and one coin is selected at random from the jar. The random variable T represents the total of the values of the 3 coins selected.

(a) Find the sampling distribution of T.

(7)

(b) Find the sampling distribution of the median of the values of the 3 selected coins.

(4)

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Question 5 continued

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Question 5 continued

(Total 11 marks)

Q5

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6. A machine is designed to cut bolts. The length, in millimetres, of each bolt is uniformly distributed over the interval [39.5, 40.5]

Bolts longer than 40.45 millimetres cannot be used.

One bolt is selected at random.

(a) Find the probability that the bolt **can** be used.

(1)

In a random sample of 10 bolts,

(b) find the probability that at least 9 bolts can be used.

(3)

In a random sample of 120 bolts,

(c) find, using a suitable approximation, the probability that more than 117 bolts can be used.

(4)

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Question 6 continued

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Question 6 continued

Q6

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7. Jonathan believes that the time, in minutes, people spend queuing for lunch at a particular fast food restaurant can be modelled using the continuous random variable X, which has cumulative distribution function

$$\mathrm{F}(x) = \begin{cases} 0 & x < 0 \\ \dfrac{1}{3125}(25x^4 - 4x^5) & 0 \leqslant x \leqslant 5 \\ 1 & x > 5 \end{cases}$$

(a) Find the mode of X.

(4)

(b) Show that the upper quartile of X is 4.0 to one decimal place.

(3)

Olivia believes that this model overestimates the proportion of queuing times that are more than 4.0 minutes. She takes a random sample of 25 queuing times and finds that 3 of them are more than 4.0 minutes.

Assuming that the model gives a probability of 0.25 that the queuing time is more than 4.0 minutes,

(c) use a suitable test, at the 5% level of significance, to assess Olivia's belief. State your hypotheses clearly.

(5)

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Question 7 continued

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Question 7 continued

Q7

(Total 12 marks)

TOTAL FOR PAPER: 75 MARKS

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